



High-Resolution Imagery for Disaster Mapping

Unmanned Systems & Image Processing Basics

Module 04 - Topic #1

Instructor: Dr. Giorgos SFIKAS (Assistant Professor, University of West Attica)

DIGHAZ is implemented by:



University of Alicante
Cluster Lead
SPAIN

The University
of West Attica
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Unmanned Systems & Image Processing Basics

Introduction to the module

- Module title: *High-Resolution Imagery for Disaster Mapping*

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Introduction to the module

- Module title: *High-Resolution Imagery for Disaster Mapping*
- Key axes in our lectures
 - Axis 1: *High-Resolution Imagery*

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Introduction to the module

- Module title: *High-Resolution Imagery for Disaster Mapping*
- Key axes in our lectures
 - Axis 1: *High-Resolution Imagery*
 - Axis 2: *Disaster Mapping*

Learning Objectives

By the end of this lecture you will be able to:

01 Understand UAV Regulatory Frameworks

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- 04 Analyze Digital Image Representation

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Topics Covered

- Aviation Regulations & Disaster Applications
- Consumer Drones versus Professional Enterprise Systems
- Impact of Unmanned Systems and Payload combinations
- Basics of Digital Image Processing

Estimated Duration
90 minutes lecture + 45 min exercise

Unmanned Systems & Image Processing Basics

Introduction to lecture topic

- Unmanned Aerial Vehicles (UAVs) : A paradigm shift

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Introduction to lecture topic

- Unmanned Aerial Vehicles (UAVs) : **A paradigm shift**
 - Reactive and proactive character of their use

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 - Regulatory frameworks, typologies, payloads

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Introduction to lecture topic

- Unmanned Aerial Vehicles (UAVs) : **A paradigm shift**
 - Reactive and proactive character of their use
 - Regulatory frameworks, typologies, payloads
 - Key to understanding strengths&limitations: A discussion of digital image processing fundamentals

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Aviation Regulations & Disaster Applications

- Operating UAVs & Regulatory framework

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Aviation Regulations & Disaster Applications

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- In the EU: The «EASA» establishes a unified framework

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Aviation Regulations & Disaster Applications

- Operating UAVs & Regulatory framework
- In the EU: The «EASA» establishes a unified framework
- As of 1/1/2024, all drones must bear a C-class id label

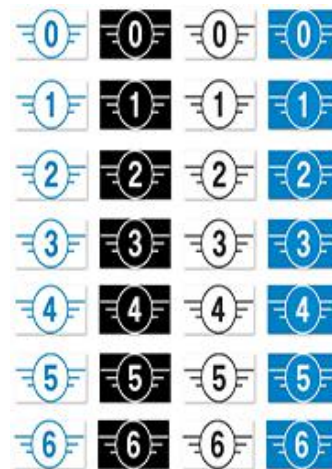


Figure 1: The drone C-Class identification labels (CILs) include the symbols for all drones classes (C0 to C6) in a format based on the EU 'Drone Regulation' (Commission Delegated Regulation (EU) 2019/945), as amended by Commission Delegated Regulation (EU) 2020/1058.

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Aviation Regulations & Disaster Applications

- Operating UAVs & Regulatory framework
- In the EU: The «EASA» establishes a unified framework
- As of 1/1/2024, all drones must bear a C-class id label
- Drones on the market before 2024: *Restricted legacy category*

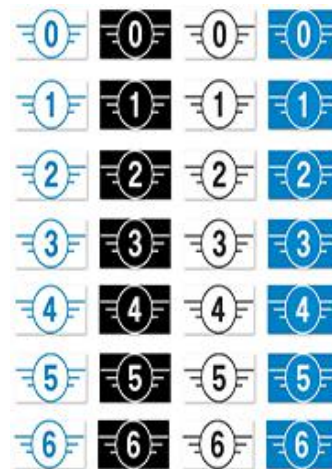


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Aviation Regulations & Disaster Applications

- Operation categorizations: Open, Specific, Certified

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- Operation categorizations: Open, Specific, Certified
- «Specific»: Authorization from National Aviation Authorities (NAAs) requirements
 - specific operations risk assessment (SORA)
 - pre-defined risk assessments (PDRA)

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- «Certified»: European Standard Scenarios
 - STS-01

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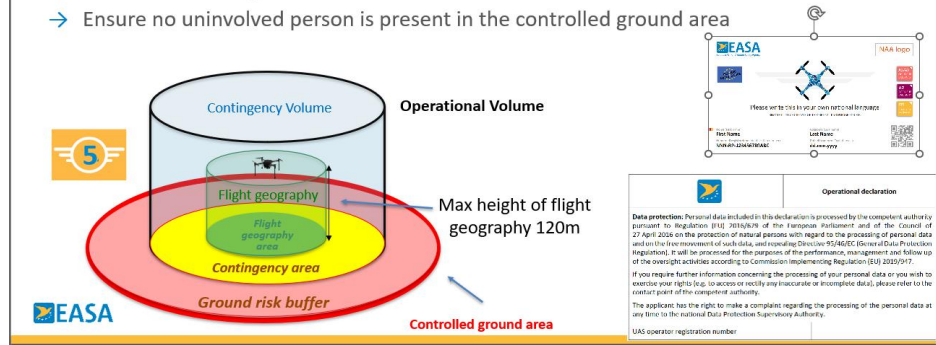
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European Standard Scenario STS - 01

Regulation 2019/947 – Appendix 1 – Chapter 1

- VLOS, below 120m (also in urban environment)
- with a UAS bearing a C5 Class identification label
- Ensure no uninvolved person is present in the controlled ground area



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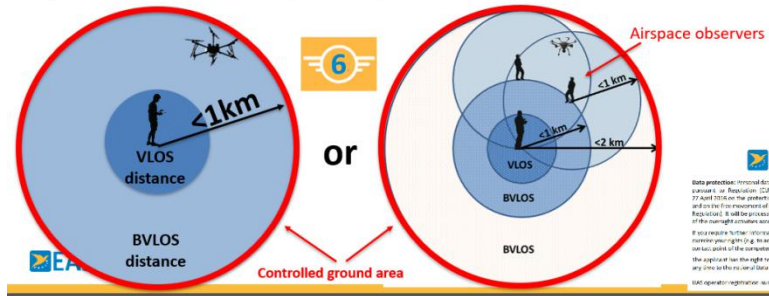
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 - STS-01
 - STS-02

European Standard Scenario STS - 02

Regulation 2019/947 – Appendix 1 – Chapter 2

- BVLOS (up to 1km distance or 2 km if airspace observer is used),
- below 120m (over sparsely populated area)
- with a UAS bearing a C6 Class identification label
- Ensure no uninvolved person is present in the controlled ground area



Operational declaration

Date protection: Personal data articulated in this declaration is processed by the competent authority pursuant to Regulation (EU) 2019/947 of the European Parliament and of the Council of 27 June 2019 on the operation of unmanned aircraft systems and on the introduction of a regulatory framework for the operation of unmanned aircraft systems, and on the introduction of a regulatory framework for the operation of unmanned aircraft systems.

If you require further information on the processing of your personal data or wish to exercise your rights (e.g. to access or rectify any inaccurate or incomplete data), please refer to the data protection officer of the competent authority.

The applicant has the right to make a complaint regarding the processing of the personal data of any one of the data controller's data processing activities.

UAS registration number:

Aviation Regulations & Disaster Applications

- Mandate for Remote ID

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Aviation Regulations & Disaster Applications

- Mandate for Remote ID
- «Certified» category for missions -- two levels of certification

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 - NAA CoAs

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Aviation Regulations & Disaster Applications

Regulatory Category	Risk Profile	Key Requirements and Authorizations	Disaster Management Applicability
Open	Low	Visual Line of Sight (VLOS) only; C-class labels mandatory; strict altitude limits; remote ID required.	Initial, localized reconnaissance; minor incident monitoring outside urban centers.
Specific	Medium	BVLOS permitted; NAA authorization via SORA, PDRA, or STS required.	Extensive SAR missions; large-scale hazard mapping; complex urban damage assessment.
Certified	High	EASA Type Certificates (TCs) and NAA Certificates of Airworthiness (CoA) required.	Heavy cargo transport (e.g., heavy medical supplies); potential autonomous passenger evacuation.

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Consumer Drones versus Professional Enterprise Systems

- Selection of UAV:
 - Depends on needs

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Consumer Drones versus Professional Enterprise Systems

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Fixed-Wing UAV	Multicopter UAV (Light)	Multicopter UAV (Heavy)
Hi-Target V100	DJI Phantom 4 Pro	DJI Matrice 350
Maximum Flight Time: 150 min	Maximum Flight Time: 30 min	Maximum Flight Time: 55 min
Maximum Flight Speed: 30 m/s	Maximum Flight Speed: 20 m/s	Maximum Flight Speed: 23 m/s
Maximum Payload: 2 kg	Aircraft Weight: 1.388 kg	Maximum Payload: 2.7 kg
Maximum Flight altitude: 6000 m	Maximum Flight altitude: 6000 m	Maximum Flight altitude: 7000 m
Adaptability to Work Environment: Open Areas	Adaptability to Work Environment: Complex Terrain	
		

Unmanned Systems & Image Processing Basics

Impact of Unmanned Systems and Payload combinations

- Impact on unmanned systems

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Impact of Unmanned Systems and Payload combinations

- Impact on unmanned systems
- Problems with ground-based response

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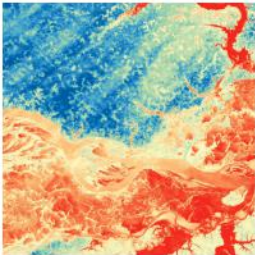
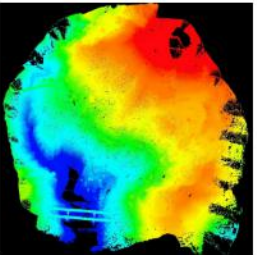
Impact of Unmanned Systems and Payload combinations

- Impact on unmanned systems
- Problems with ground-based response
- Risk mitigation with drones

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Impact of Unmanned Systems and Payload combinations

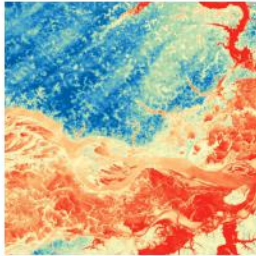
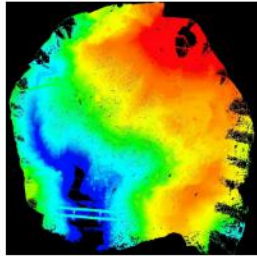
- Visible spectrum (RGB) cameras
- Thermal cameras

Visible Light Camera	Thermal Infrared Camera	Multispectral Camera	LiDAR
Main Application: High-resolution color imaging	Main Application: Infrared radiation and temperature detection	Main Application: Multispectral vegetation classification	Main Application: Laser ranging, 3D modeling
Drawbacks: Dependent on daylight conditions	Drawbacks: Lower spatial resolution	Drawbacks: Limited spatial resolution	Drawbacks: Atmospheric interference, high cost
	Specific influencing factors: Image contrast decreases by 40% under rainy conditions, and more than 50% of texture information is lost under foggy conditions.		Specific influencing factors: heavy-fog conditions reduce point cloud density by approximately 30%, while rainy/overcast conditions increase the error by 5–10 cm.
			

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Impact of Unmanned Systems and Payload combinations

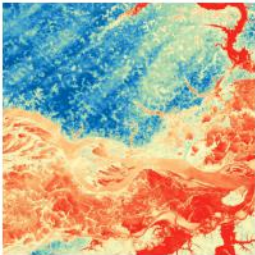
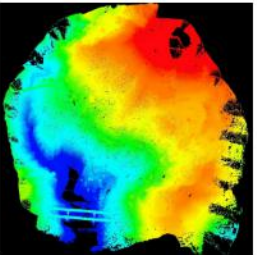
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- LiDAR sensors
- Specialized devices

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Unmanned Systems & Image Processing Basics

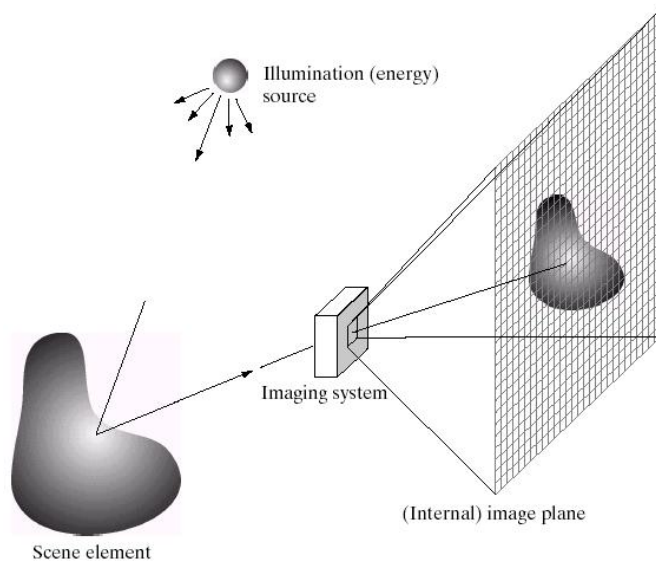
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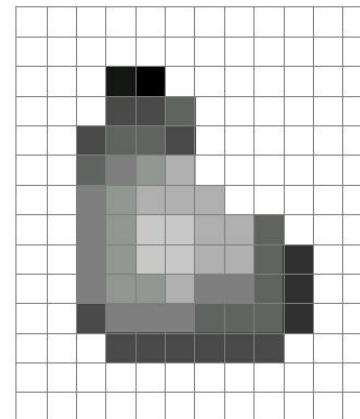
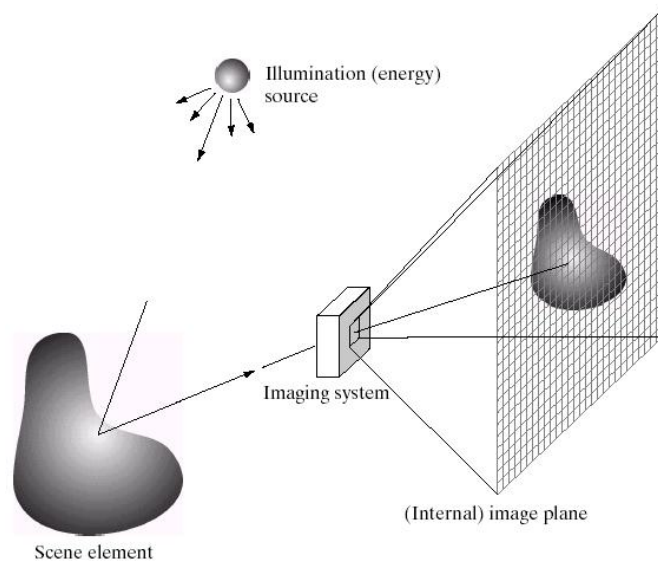
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Basics of Digital Image Processing



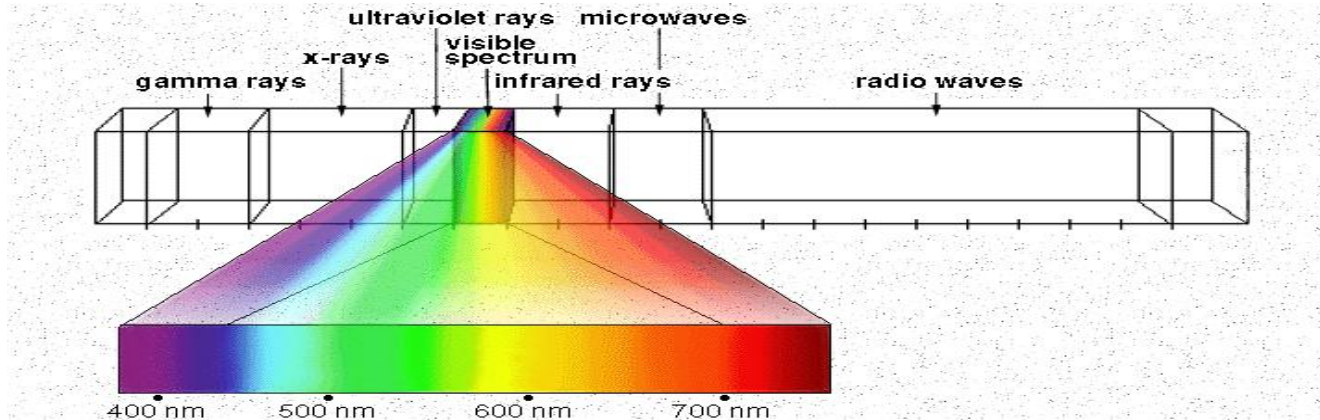
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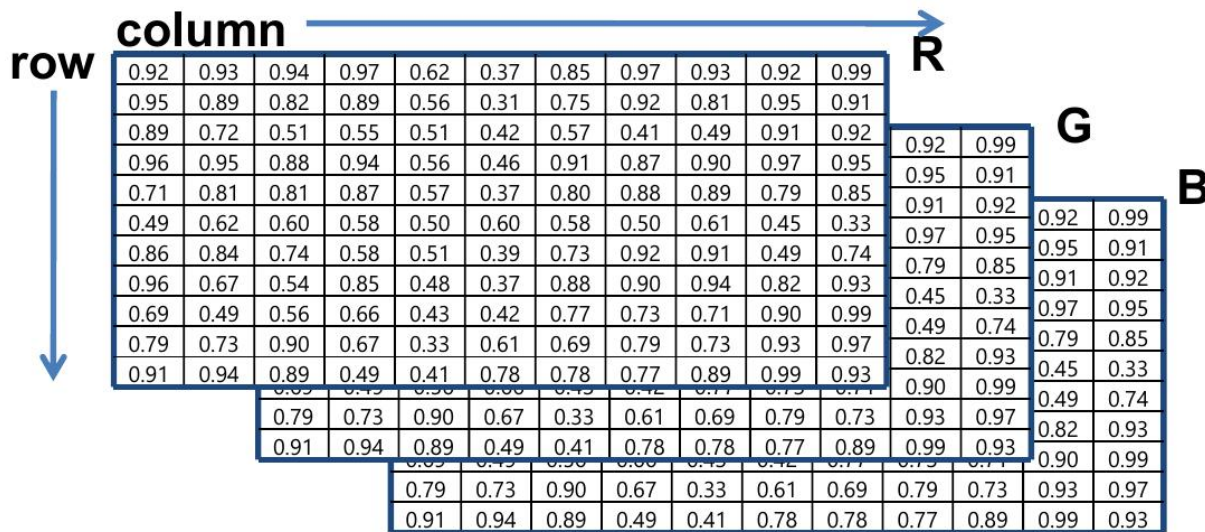
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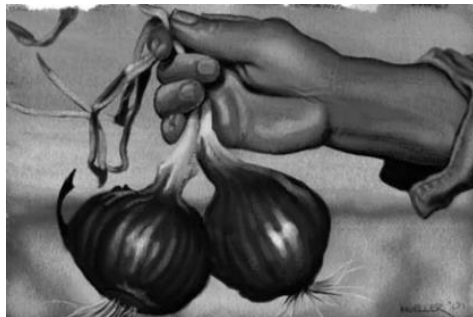
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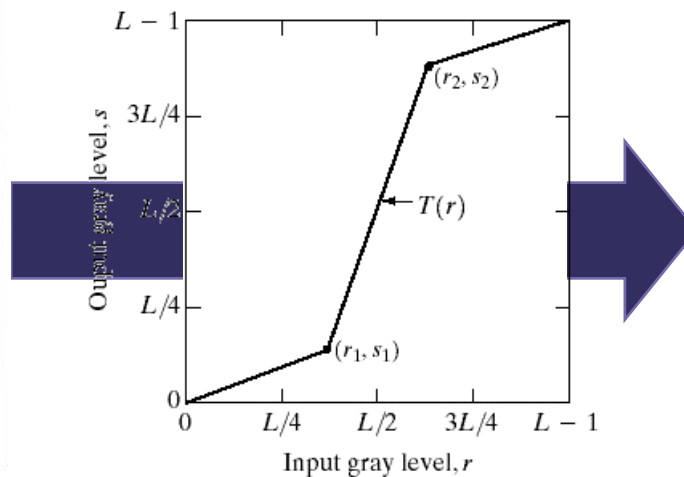
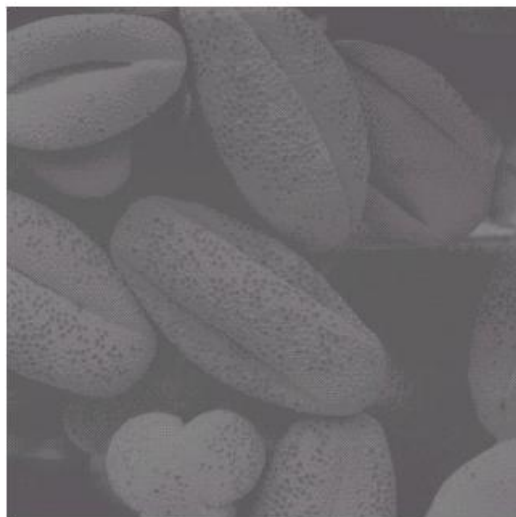
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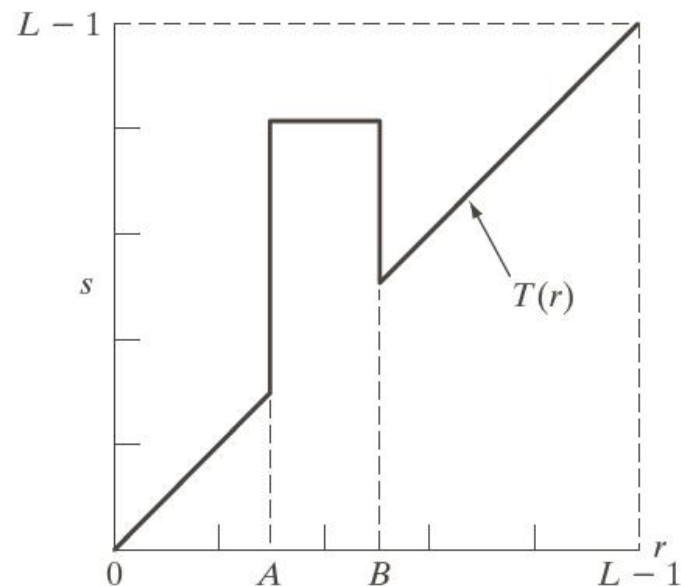
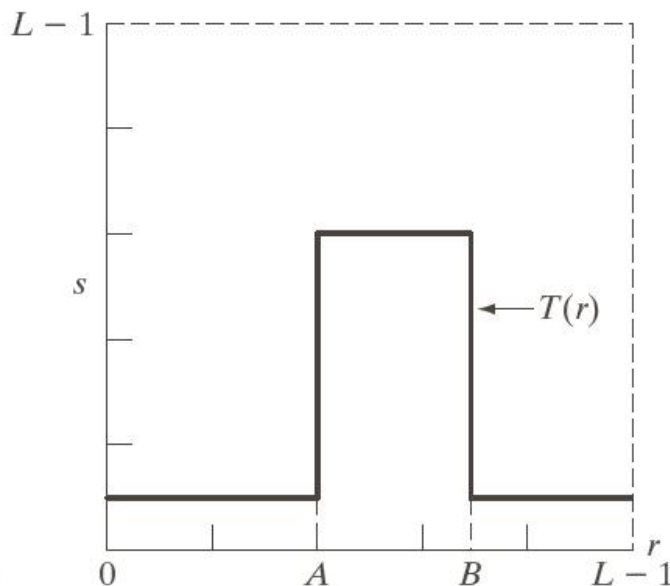


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Basics of Digital Image Processing

a b

FIGURE 3.11 (a) This transformation highlights intensity range $[A, B]$ and reduces all other intensities to a lower level. (b) This transformation highlights range $[A, B]$ and preserves all other intensity levels.



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Discussion questions

1. What are some of the most important types of sensors that UAVs can carry?
2. Why is digital imaging important for disaster mapping & response?
3. Search (web, LLM) for research conferences & journals on the fields of Digital Image Processing, Computer Vision, Photogrammetry. Discuss the overlap between these fields, and between each of them to the field of Machine Learning.

Sneak peak at your exercise

Example

Opening a digital image and processing it

See also:

GitHub repo / Notebook link

```
# Load standard imaging libraries in Python
import numpy as np
import matplotlib.pyplot as plt
from PIL import Image

# Load a test image
import skimage.data as fixtures
A = fixtures.camera()
# Visualize it
plt.imshow(A, cmap="gray")
plt.show()
# We can manipulate the image content by manipulating the
numpy array
# For example:
A2 = np.copy(A)
A2[300:400, 300:400] = 0
plt.imshow(A2, cmap="gray")
plt.show()
```

Thank you!

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